THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Earth, Moon, and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** rotation	causes day and night
2	**(c)** revolution	takes one year
3	**(a)** tilted axis and revolution	tilt of 23.5°
4	**(a)** 22 December	winter solstice
5	**(d)** natural satellite	of the Earth
6	**(c)** new moon (Amavasya)	dark side faces Earth
7	**(a)** Moon	blocks sunlight
8	**(c)** umbra	total darkness
9	**(b)** from anywhere on the night side of Earth	safe to view
10	**(a)** Earth's rotation	daily motion

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Spinning on its axis (1); day and night (1).	1+1
12	Motion around the Sun (1); one year (1).	1+1
13	A year of 366 days (1); to account for extra 0.25 day yearly (1).	1+1
14	Earth's axis is tilted (1); and it revolves around the Sun (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Earth is a sphere lit by the Sun (1); half faces Sun (1); rotation changes sides (1).	1+1+1
16	Tilt changes direction of sunlight (1.5); more direct rays give summer (1.5).	1½+1½
17	Hemisphere tilted towards Sun gets more daylight hours (3).	3
	OR: New moon, waxing crescent (1); first quarter, full moon (1); waning phases back to new moon (1).	1+1+1
18	Moon between Sun and Earth (1); on new moon day (1); shadow falls on Earth (1).	1+1+1
	OR: Earth between Sun and Moon (1); on full moon day (1); Earth's shadow on Moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Spin; orbit (2). (ii) 24 hours; 365¼ days (2). (iii) Day-night; seasons (1).	2+2+1
	OR: (i) Earth's tilt and revolution (1). (ii) Changes sunlight angle (1). (iii) Solstices (1). (iv) Opposite seasons (2).	1+1+1+2
20	(i) Tilt, revolution (2). (ii) Definitions (2). (iii) Not distance from Sun (1).	2+2+1
	OR: (i) Relative positions (1). (ii) Waxing (1). (iii) Purnima (1). (iv) Amavasya (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 11: **Earth, Moon, and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 24 hours	one day
2	**(d)** 365¼ days	one year
3	**(d)** 23.5°	from vertical to orbit plane
4	**(a)** equinoxes	21 March and 23 September
5	**(c)** reflects sunlight	non-luminous
6	**(a)** 29.5 days	synodic month
7	**(b)** new moon day	alignment
8	**(a)** eyes	retina
9	**(b)** Moon's orbit is tilted to Earth's orbit	alignment is rare
10	**(c)** near the extension of Earth's axis	Dhruva Tara

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Days with longest or shortest daylight (2).	2
12	Days when day and night are equal (2).	2
13	We see different portions of its lit half (2).	2
14	Waxing: lit part increasing (1); waning: decreasing (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solar: Moon blocks Sun, new moon (1.5); lunar: Earth's shadow on Moon, full moon (1.5).	1½+1½
16	Use certified filters (1.5); pinhole projection (1.5).	1½+1½
17	Earth revolves (1.5); we face different parts of space (1.5).	1½+1½
	OR: It is a rocky body (1.5); shines by reflecting sunlight (1.5).	1½+1½
18	Year divisible by 4 (1.5); 2024, 2028 (1.5).	1½+1½
	OR: One side always day and hot (1.5); other always night and cold (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sun–Moon–Earth in line (2). (ii) New moon (1). (iii) Filters (2).	2+1+2
	OR: (i) Earth's shadow on Moon (2). (ii) Whole night side sees (1). (iii) Differences (2).	2+1+2
20	(i) New moon (1). (ii) Sun, Moon, Earth (1). (iii) Filters/projection (1). (iv) Eye damage (2).	1+1+1+2
	OR: (i) Tilted orbit (2). (ii) A few times a year (1). (iii) Study of Sun's corona (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.